# Exit Ticket · reflect and retrieve

## Exit Ticket

O(log n) means execution time is literally log n seconds.

## Use key vocabulary from memory

1. Explain time complexity in your own words.

2. Explain O(n) in your own words.

3. Explain O(log n) in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 189 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Search and sort: comparisons and Big O

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What property does binary search need that linear search does not?

### Q2

What time complexity is typical of merge sort?

### Q3

For n=5, how many times does a single for-loop from 0 to n-1 run?

### Q4

A nested loop runs 4 times outside and 4 times inside. How many inner operations occur?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Count what grows — Big O describes how work increases with input size, not the exact milliseconds on one computer.

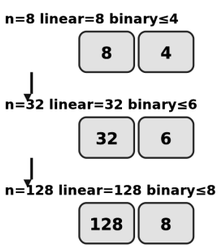
## Core idea 2

Compare search — Linear search can take n checks; binary search on sorted input roughly halves the remaining candidates each step.

## Core idea 3

Compare sorting — A simple nested comparison loop can be O(n²), whereas merge sort is O(n log n).

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Is binary search always better than linear search? Give one counterexample.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Name the complexity of a single loop visiting n values.

## Question 2

An array has 128 ordered items. Compare linear and binary worst-case checks.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

time complexity — How an algorithm's work grows with input size.

O(n) — Work proportional to input count.

O(log n) — Work proportional to the number of halvings.

O($n^2$) — Quadratic growth, often nested loops.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

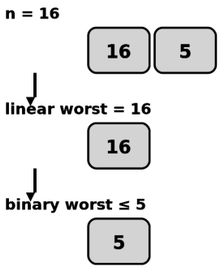
For 16 values, compare maximum comparisons of linear and binary search.

## Follow the worked method

1. Predict first — Assume one successful or unsuccessful lookup in sorted data.

2. Model solution — Linear search can inspect all 16 elements; binary search checks at most 5 midpoints using this inclusive implementation.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first
What happens when you double 32 to 64?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.